

Domestic operators such as Ishan Technologies, RackBank, NeevCloud, Colt DCS and others are also expanding data centre capacity across the country. Separately, there are also enterprise partnerships that are being developed. In May, global ride-hailing major Uber entered into a pact with the Adani Group to set up its first data centre in India.

According to Sambit Patra, Partner and India Head of Energy & Natural Resources at Bain & Company, India has added close to 1 gigawatt of data-centre capacity over the last five years and is now racing towards achieving four to six times that capacity by 2030.

Data centre capacity, measured in watts, houses processors, ultra-fast networking, storage and massive power and cooling systems. As AI adoption explodes across sectors, including banking, healthcare, telecom, manufacturing, retail, and government services, more of these centres are being set up to support these services.

Data centres are not new to India. They were first established in 2000 to help power websites, banking systems, enterprise software, payments, cloud storage, and so on. These conventional data centres run on standard central processing units (CPUs). However, AI relies heavily on graphic processing units (GPUs) that process thousands of calculations simultaneously to train and run large models. These chips are significantly more power-intensive and generate far greater heat, requiring denser racks, advanced cooling systems and substantially higher electricity and water use than traditional facilities.

“The next phase of AI infrastructure growth will be defined not just by access to compute hardware, but by the ability to se-

cure large-scale power ecosystems capable of sustaining continuous high-density AI workloads,” says Narendra Sen, founder and CEO RackBank and NeevCloud. Bengaluru-based NeevCloud is a sovereign AI cloud company, while RackBank is a data centre services provider, both founded by Sen, a technology entrepreneur.

AI campuses that once consumed 20-40 megawatts (MWs) of power are now being planned at 100-300 MW and sometimes even at a gigawatt scale. Rack densities (a measure of electrical power consumed, and heat generated) have jumped from levels of 5-10 kilowatts to as high as 150 kilowatts per rack, as companies cram more GPUs into facilities designed to train large language models.

The infrastructure is key to AI and promises jobs, cloud sovereignty, low-latency digital services and the computing muscle that India needs to compete globally. According to an IBM-India AI study released in May, AI could add \$500 billion to India’s economy by 2030.

“India’s digital public infrastructure has created one of the strongest foundations for innovation globally. The next opportunity is to build on that momentum by enabling AI infrastructure at scale,” says Himani Agrawal, COO, Microsoft India and South Asia.

Microsoft has committed \$17.5 billion over four years toward AI and cloud expansion in India, including a hyperscale region in Hyderabad. Amazon plans to invest over ₹1 lakh crore in cloud infrastructure by 2030. Google, too, is building massive AI infrastructure projects, including a proposed facility in Andhra Pradesh.

Despite the promises, the infrastructure powering India’s AI ambitions is also emerging as a for-

MAKING AI RESPONSIBLE

○
Data centres consume enormous amounts of power, water

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Power demand from these centres to double from 0.5-1% to 1-2% of total power demand by 2030

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Water consumption to increase from 150 billion litres in 2025 to 358 billion litres by 2020

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Strategy should be to locate the centres at places where renewable power is easily available

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Need to scout for alternatives to water for cooling data centres; split demand between large and mini-data centres; select sites most optimal for large and small centres

